QUASI-PROJECTIVITY OF A REGULAR TWO DIMENSIONAL SCHEME OVER A NOETHERIAN BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the application of Chow's lemma, the extension of the rational map along blowups in closed points, and the descent of quasi-projectivity along the chain of blowups.

Editorial note. The label lemma-regular-dim-2-quasi-projective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-regular-dim-2-quasi-projective –start 5049 –end 5080 –local-id 28105 returned RESULT=OK (L28105 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Chow's lemma. Cohomology of Schemes, the Noetherian version of Chow's lemma. Produces a proper modification of the source which is H-quasi-projective over the base and an isomorphism over a dense open.
- Extending a rational map along a sequence of blowups in closed points. Same chapter, the lemma producing a sequence of blowups in closed points lifting a rational map into a modification. Provides the chain of blowups through which the rational map to the H-quasi-projective modification is defined.
- Permanence of projectivity and quasi-projectivity, and contraction of the last blowing up. Morphisms, the lemmas on the composition and permanence of projective and quasi-projective morphisms, together with the lemma on contracting the last blowup when the first blowup is quasi-projective and the uniqueness of contractions. Transfers quasi-projectivity from the last blowup back down the chain to the original scheme.

Source reference. Original lemma 0200: sources/stacks-chow/tags.txt, source lines 2617–2617. Prerequisite pointer invoked at proof line(s) 5058; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02NS: sources/stacks-chow/tags.txt, source lines 3491–3491. Prerequisite pointer invoked at proof line(s) 5068; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C2N: sources/stacks-resolve/pinned-original-source.tex, source lines 4986–4986. Prerequisite pointer invoked at proof line(s) 5076; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C5K: sources/stacks-resolve/pinned-original-source.tex, source lines 4592–4592. Prerequisite pointer invoked at proof line(s) 5077; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C5H: sources/stacks-resolve/pinned-original-source.tex, source lines 1182–1182. Prerequisite pointer invoked at proof line(s) 5061; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C4P: sources/stacks-chow/tags.txt, source lines 15234–15234. Prerequisite pointer invoked at proof line(s) 5069; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C4M: sources/stacks-chow/tags.txt, source lines 15232–15232. Prerequisite pointer invoked at proof line(s) 5073; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C4Q: sources/stacks-chow/tags.txt, source lines 15235–15235. Prerequisite pointer invoked at proof line(s) 5071; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07RL: sources/stacks-chow/tags.txt, source lines 9871–9871. Prerequisite pointer invoked at proof line(s) 5075; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a Noetherian scheme. Let $f : X \to S$ be a separated morphism of finite type with X regular of dimension 2. Then X is quasi-projective over S.*

Proof. By Chow's lemma (Cohomology of Schemes, Lemma 0200) there exists a proper morphism $\pi : X' \to X$ which is an isomorphism over a dense open $U \subset X$ such that $X' \to S$ is H-quasi-projective. By Lemma 0C5H there exists a sequence of blowups in closed points

$$X_n \to \ldots \to X_1 \to X_0 = X$$

and an S-morphism $X_n \to X'$ extending the rational map $U \to X'$. Observe that $X_n \to X$ is projective by Divisors, Lemma 02NS and Morphisms, Lemma 0C4P. This implies that $X_n \to X'$ is projective by Morphisms, Lemma 0C4Q. Hence $X_n \to S$ is quasi-projective by Morphisms, Lemma 0C4M (and the fact that a projective morphism is quasi-projective, see Morphisms, Lemma 07RL). By

Lemma 0C2N (and uniqueness of contractions Lemma 0C5K) we conclude that $X_{n-1}, \ldots, X_0 = X$ are quasi-projective over S as desired. □